

Day 2 — The Carry Case

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will correctly handle the carry case in 2-digit $\times 11$ (where the digit sum ≥ 10), including problems like $87 \times 11 = 957$, in under 5 seconds.

Materials

- Whiteboard.
- Printed worksheet of 12 problems, mixing no-carry and carry cases.
- Stopwatch.

Lesson flow

Warm-up (5 min)

- Neighbour-sum drill: teacher calls a 2-digit number with digit-sum < 10 ; class chorus the $\times 11$ product.
- “12 times 11!” $\rightarrow$ “132!” — “34 times 11!” $\rightarrow$ “374!” — 10 in a minute.

Recall (3 min)

- Yesterday’s problem: 73×11 . $7 + 3 = 10$. Why was this awkward?
- “You can’t put 10 in the middle of a 3-digit number. So what do we do? Carry the 1 — exactly like in standard addition.”

Core (15 min) — The carry case

1. Demonstrate 73×11 fully:

Step 1: Split: 7 3

Step 2: Add: 7 + 3 = 10

Step 3: Write the units digit of the sum (0) in the middle: 7
0 3

Step 4: Carry the tens digit of the sum (1) into the leftmost digit: $7 + 1 = 8$.

Step 5: Answer: 8 0 3 $\rightarrow$ 803.

Check: $73 \times 11 = 73 \times 10 + 73 = 730 + 73 = \mathbf{803}$. ✓

2. Try 87×11 :

8 7

8 + 7 = 15

Middle = 5; carry 1 into the 8 $\rightarrow$ 9.

Answer: 9 5 7 $\rightarrow$ 957.

Check: $87 \times 11 = 870 + 87 = 957$. ✓

3. **Try 99×11 :**

$$\begin{array}{r} 9 \\ 9 \end{array} \begin{array}{r} 9 \\ + 9 \end{array} = 18$$

Middle = 8; carry 1 into the 9 $\rightarrow$ 10.

But 10 isn't a single digit – so it becomes a "1" carried further: leftmost = 0, with a 1 carried to the left of that.

Answer: 1 0 8 9 $\rightarrow$ 1089.

Check: $99 \times 11 = 990 + 99 = 1089$. ✓ (4-digit answer because the carry cascaded.)

4. **The full procedure** (boxed on the board):

For any 2-digit number $ab \times 11$:

1. Compute $a + b$.
2. If the sum is < 10 : answer is $a (a+b) b$.
3. If the sum is ≥ 10 : middle digit = units digit of sum; add 1 to a .
4. If the new a is ≥ 10 (only happens at 99): cascade the carry.

Activity (17 min) — Mixed practice

- Worksheet of 12 problems, half no-carry, half carry:

21×11	35×11	47×11	58×11
62×11	74×11	83×11	89×11
91×11	99×11	18×11	27×11

- Round 1 (8 min): solo, untimed. Self-check with long multiplication on any 3 of them.
- Round 2 (8 min): paired. Partner calls a problem; you answer aloud. Goal: under 5 seconds per problem.

Wrap-up (5 min)

- Quick chorus: “*What does the trick break into?*” — Split, add, drop (and carry if needed).
- Tease Day 3: “*Tomorrow we PROVE why this works. You’ll write the algebra yourself.*”

Student-facing content

The $\times 11$ Trick (with carry)

1. Split the digits.
2. Add them.
3. **If the sum < 10 :** drop it in the middle. Done.
4. **If the sum ≥ 10 :** keep the units digit of the sum as the middle; add 1 to the leftmost digit.

Examples: - 45×11 : $4+5=9 \rightarrow 495 = 495$ - 87×11 : $8+7=15 \rightarrow 8(1)57 \rightarrow 957 = 957$ -

99×11 : $9+9=18 \rightarrow 9(1)89 \rightarrow \text{cascade} \rightarrow 1089 = 1089$

Homework

- 8 mental $\times 11$ problems (carry case): 76, 84, 85, 88, 93, 96, 67, 78 — multiplied by 11. Write the answers.
- Time yourself. Goal: under 30 seconds total.

Differentiation

- **Tier up:** Find a 2-digit number where the trick gives a 4-digit answer (i.e. $99 \times 11 = 1089$). Can you find any others, or is 99 unique? Justify.
- **Tier down:** Solve only the no-carry problems on the worksheet today. We'll cement carries on Day 4.

Teacher notes

- The most common error: students remember to write the units digit of the middle sum but **forget** to add the carry into the leftmost digit. Watch for “ $73 \times 11 = 703$ ” with no carry added — this is a classic mistake.
- The 99×11 case (and the cascading carry) genuinely surprises students. Lean into it — it's evidence the algebra is doing real work, not arithmetic by coincidence.
- If a student asks “why does this work?” — that's Day 3. Park it. Honour the curiosity by writing the question on a board sticky.

Citation(s) used in this lesson

- Tirthaji (1965), sūtra 1 (*Ekādhikena Pūrveṇa*) — the technique name.
- *Vedic Mathematics Batch1.pdf* (Vedic Cultural Center) — worked examples of the carry case.